

Estefania Marmolejo Sandoval

Nayarit, Mexico

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SUMMARY

AI & Data Engineer driven by innovation and purpose. Experienced in designing intelligent data pipelines and cloud-native architectures that turn complex data into actionable insights. Passionate about applying AI responsibly to enhance healthcare, accessibility, and real-world decision-making.

SKILLS

Python, Java, C#, JavaScript, SQL, PostgreSQL, MongoDB, AWS, Azure, GCP, Oracle Cloud, ETL pipelines, Data Engineering, Machine Learning, Generative AI, Llama ecosystem, Applied AI tools, OT/IT Security, OpenShift, REST APIs, CI/CD, React Native, Alembic / SQLAlchemy, Postman, Docker, Figma, Git

WORK EXPERIENCE

Mxmart Solutions Aug 2025 - Present
AI & Data Engineer Jalisco, Mexico

- Designed and deployed a serverless architecture on Azure for automated document information extraction, utilizing Python, Azure OpenAI, and Document Intelligence models.
- Built and optimized AI pipelines for data extraction, validation, and persistence using PostgreSQL, SQLAlchemy, and Alembic.
- Developed secure REST endpoints for the integration of document processing systems, ensuring robust security measures for data management.
- Created dynamic Excel report generation tools and implemented AI usage and cost tracking per user, optimizing model deployment efficiency.
- Integrated a Next.js, TypeScript, and Tailwind CSS frontend, incorporating administrative modules and real-time notifications with Cloud Pub/Sub and WebSockets to enhance system performance and user experience.

Mxmart Solutions Jan 2025 - Aug 2025
Cloud & Data Engineer Jalisco, Mexico

- Co-designed and deployed a hybrid cloud fleet management system, leveraging AWS, Azure, and GCP to ensure scalable, secure, and high-performance data flows across logistics operations.
- Developed SQL/Python data pipelines for data extraction, transformation, and loading (ETL), optimizing the flow of logistics data and integrating it into the cloud infrastructure.
- Built real-time dashboards using Power BI/Grafana to provide key performance indicators (KPIs) for fleet operations, improving decision-making and operational efficiency.
- Collaborated with cross-functional teams to ensure seamless data integration between cloud platforms and on-premise systems, optimizing overall system performance and data accessibility.

ECN Automation Aug 2024 - Jan 2025
Internship Sonora, Mexico

- Designed and deployed a scalable industrial monitoring system to integrate IoT sensors and provide real-time data for critical system monitoring, ensuring high availability and system resilience.
- Developed interactive visualizations using Grafana to display KPIs and system performance metrics, enhancing operational visibility and enabling faster, data-driven decision-making across teams.
- Collaborated with cross-functional teams to optimize the system for industrial environments, ensuring ease of integration and scalability for future system expansions.

PROJECTS

Memora – AI & Mobile App for Alzheimer’s Support May 2025 - Present
Founder & Project Lead Nayarit, Mexico

- Developing a mobile application with AI for Alzheimer’s support, focused on providing reminders for medication, appointments, and daily tasks, as well as generating summaries for caregivers. Although currently in development, the project is aimed at improving patient autonomy and easing caregiver responsibilities. The app integrates AI chatbots to enhance communication and assist in memory-related tasks, using React Native and natural language processing (NLP) technologies.

DELFIN Research Summer MMXXIII Jun 2023 - Aug 2023
Research Project Participant Cartagena, Colombia

- Collaborated in an international research project at Universidad Tecnológica de Bolívar, focused on detecting hate speech and homophobic content on social media platforms using NLP and machine learning techniques.

- Employed text classification models and sentiment analysis to identify and categorize harmful language, contributing to the development of more accurate detection algorithms.
- IX International Meeting of Research Internships DELFIN 2023**
Speaker

Jul 2023 - Jul 2023
Manizales, Colombia

- Presented the project titled "Detection of Homophobic Messages on Social Networks" as a speaker at the IX International Conference of Research Internships DELFÍN 2023, focusing on the use of NLP and machine learning techniques for detecting harmful content on social media platforms.
 - Engaged with an international audience, contributing to the academic exchange on AI ethics, content moderation, and social media impact.

EXTRACURRICULARS

- SoftServe**
Program Participant

Sep 2025 - Present

- Database Fundamentals Program – Selected participant in a professional training program focused on relational databases and SQL.
 - Linux Fundamentals Program – Hands-on training in Linux system administration, shell scripting, and server management.
- LLaMA ecosystem for AI development**
Program Participant

Aug 2025 - Sep 2025

- Completed a 14-hour Bootcamp organized by Meta and INFOTEC, focused on retrieval-augmented generation (RAG) and semantic search integration within AI workflows.
 - Gained hands-on experience in structuring RAG pipelines for efficient knowledge retrieval, and explored the application of vector stores to enhance AI-driven semantic search capabilities at scale.
 - Developed a deeper understanding of LLaMA ecosystem tools for AI development, focusing on custom knowledge integration and improving AI systems' efficiency in large-scale data processing.

EDUCATION

- Instituto Tecnológico de Tepic**
Computer Systems Engineering

May 2019 - May 2025
GPA: 92.4

CERTIFICATIONS

- IELTS Academic (Band 7.5)**

Demonstrates proficiency in the English language, with a focus on academic and professional settings. The certification covers reading, writing, listening, and speaking components, ensuring effective communication in both research and international professional environments.
- AWS Certified Cloud Practitioner**

This certification validates foundational knowledge of AWS cloud services. It covers cloud computing concepts, AWS core services, security and compliance, and cloud architecture principles. This certification enables me to design and implement cloud-based infrastructure efficiently.
- Azure Administrator Associate (AZ-104)**

Focuses on the skills required to manage Azure subscriptions, virtual networks, and storage solutions. This certification demonstrates my proficiency in managing Azure cloud resources, deploying virtual machines, configuring networks, and ensuring the security and scalability of cloud environments.
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**

This certification provides foundational knowledge of Oracle's AI tools, focusing on the integration and deployment of machine learning models, data analysis, and automation within Oracle Cloud environments. I have learned to apply AI models to build scalable systems using OCI technologies.
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**

Specializes in the application of Generative AI within Oracle Cloud services. The certification covers areas such as NLP, image generation, and creating custom AI models to enhance the development of AI-driven systems in various industries.